Requirements and service boundaries

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Make requirements observable before adding model behaviour.

Teach separate operation outcomes and unchanged-source checks with complete local fixtures.

Keep one habit: Model the trace → remove help → inspect simulation evidence → assess fresh inventory reasoning → revisit a missing-record case.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of requirements and service boundaries

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Neural network experiment defense. Give an example and a mistake that the idea helps you avoid.

Look for: Defend a network experiment using traceable **evidence**

Reference example, not the only acceptable wording: Teacher explanation: Question/criterion, data and split IDs, baseline rule, architecture, initial values and learning rates, update count/loss, checkpoints and raw **prediction**s, validation selection rule, frozen final record, limitations and revision history are acceptable items.

Pictured end goal

Explain requirements and service boundaries using the deciding step in this fictional service.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Neural network experiment defense

Use the three-session teaching cycle above, with extra time as needed. A **service boundary** states who can request what, what data may be read and which actions are outside scope. Requirements must be observable. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of requirements and service boundaries

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach knowledge, permission and observability together

> Use complete local fixtures without implying live accounts or deployed protection.

Definitions and complete classroom contract

A **service boundary** states who can request what, what data may be read and which actions are outside scope. Requirements must be observable.

The fictional student timetable service reads a public schedule and returns a meeting time or review status. It cannot edit schedules or email users. Test success and refusal before adding model behaviour.

Contract question	This classroom service
Who requests?	A fictional student; identity/authentication is not implemented here
Approved evidence	Supplied public-schedule-classroom-v1: Tuesday → 16:00
Allowed operation	Read a meeting time from that supplied schedule
Excluded operations	Move/edit meetings; send email; any other action
Read with no supplied entry	Return review status, without inventing a time

Contract question	This classroom service
Mixed request	Return a separate outcome for each operation
Observable requirements	Supported read names its source; excluded actions do not execute; schedule stays unchanged

Main trace and precise expected reasoning

Split the request before deciding each operation

OPERATION	PERMISSION AND EVIDENCE CHECK	OUTCOME
Read Tuesday time	Read allowed; Tuesday entry says 16:00	answered: Tuesday 16:00, source public-schedule-classroom-v1
Move Tuesday meeting to Wednesday	Edit is outside the contract	out_of_scope: no change executed
Read Wednesday time, a separate probe	Read allowed; Wednesday entry absent	review: no supplied entry, no guessed time

One allowed operation does not authorize another. Missing evidence and excluded operations receive different outcomes.

Ask learners to name both timetable operations before showing outcomes. Read permission plus an approved Tuesday record supports the answer; neither the record nor a model's suggestion grants edit permission. Missing Wednesday is review, while a missing required day field is invalid input. Inspect the unchanged source as well as returned statuses. "review" records unresolved work; no reviewer outcome is available.

For the generic future-case question, require a fresh case under the same contract rather than generalising from one success.

Requirements and service boundaries — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A request asks for Tuesday meeting time and to move it to Wednesday. The public schedule lists Tuesday 16:00.	The fictional student timetable service reads a public schedule and returns a meeting time or review status. It cannot edit schedules or email users. Test success and refusal before adding model behaviour.	Report Tuesday 16:00 from the schedule; refuse the edit because it is outside the service boundary.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Report Tuesday 16:00 from the schedule; refuse the edit because it is outside the **service boundary**. Qualification: A helpful answer does not grant an operation that the service contract excludes.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

A request asks for Tuesday meeting time and to move it to Wednesday. The public schedule lists Tuesday 16:00.

Report Tuesday 16:00 from the schedule; refuse the edit because it is outside the **service boundary**.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Normal, missing and excluded probes

Complete probe	Expected outcome and reason
Read Tuesday; then move it	Answer 16:00 from source; reject move; schedule unchanged
Read Wednesday, no supplied entry	review; do not invent a time
Send email	out_of_scope; no email function exists in the simulation
Unknown structured operation	out_of_scope under the explicit read-only allowlist
Read action with missing day field	Invalid structured request; complete the input before a lookup
A newer conflicting schedule appears	Current one-source exercise supplies no conflict-resolution rule; review/version the contract before extending it
Return review status	Records an unresolved case; no reviewed decision is implied

Separate museum givens and scope

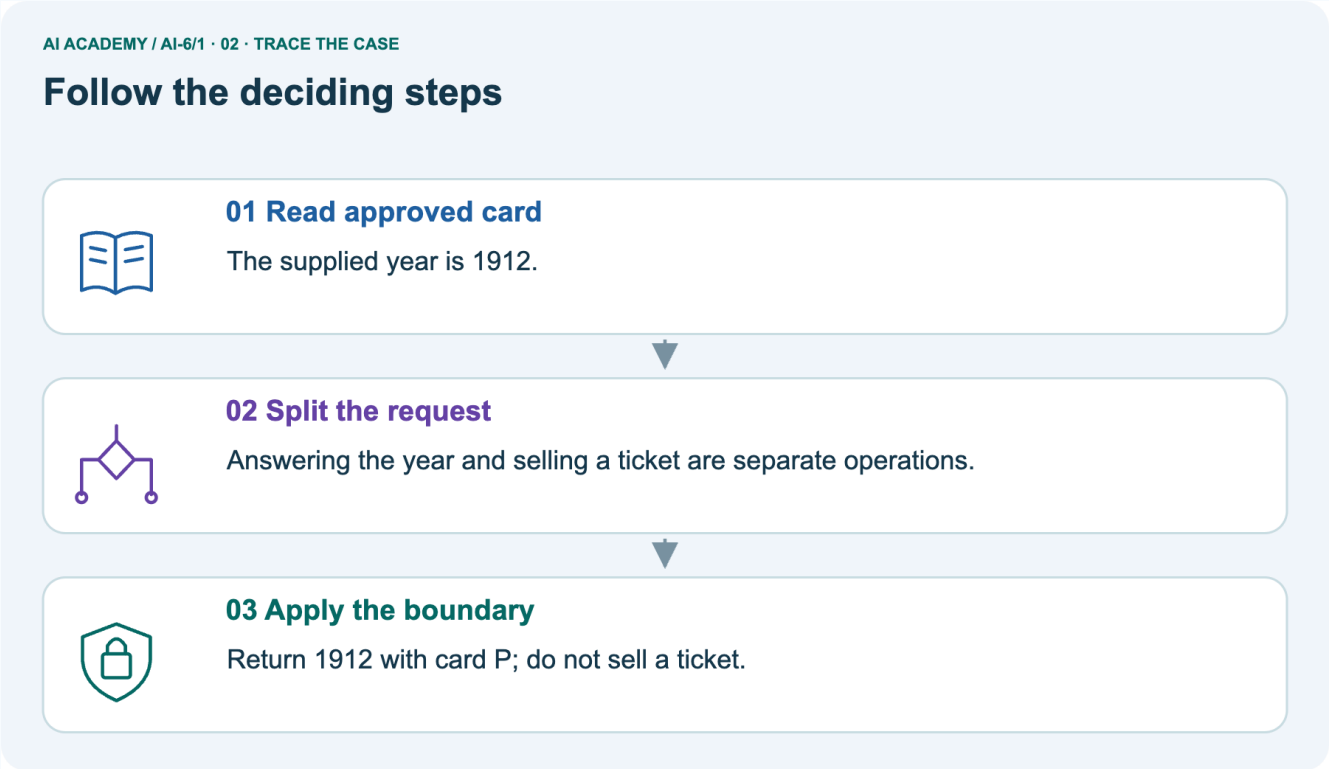
Museum fact desk. A visitor asks the opening year of exhibit P and asks the bot to sell a ticket.

Approved card P says 1912. The contract permits reading cards, not selling tickets.

The main case is a timetable; the picture is a separate museum fact desk. Approved card P supplies year 1912. The contract allows reading cards but excludes selling tickets. Return the supplied year with card P and decline the ticket sale. For Q with no approved card, the year is unsupported and needs review; selling a ticket remains excluded.

No real museum history, ticket system or human review outcome has been checked in this example.

Museum trace and missing-card key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read approved card	The supplied year is 1912.
2. Split the request	Answering the year and selling a ticket are separate operations.
3. Apply the boundary	Return 1912 with card P; do not sell a ticket.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **service boundary** states permitted work; useful information does not grant extra operations.

An approved card can be outdated; this trace does not independently verify museum history.

Change: Replace P with Q, which has no approved card.

Prediction: The year is unsupported, and the ticket sale is still excluded.

Reason: Knowledge and permission are separate requirements.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Service boundary	The declared inputs, data and operations of a service
Acceptance criterion	An observable requirement
Out of scope	Not authorised by this service contract

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

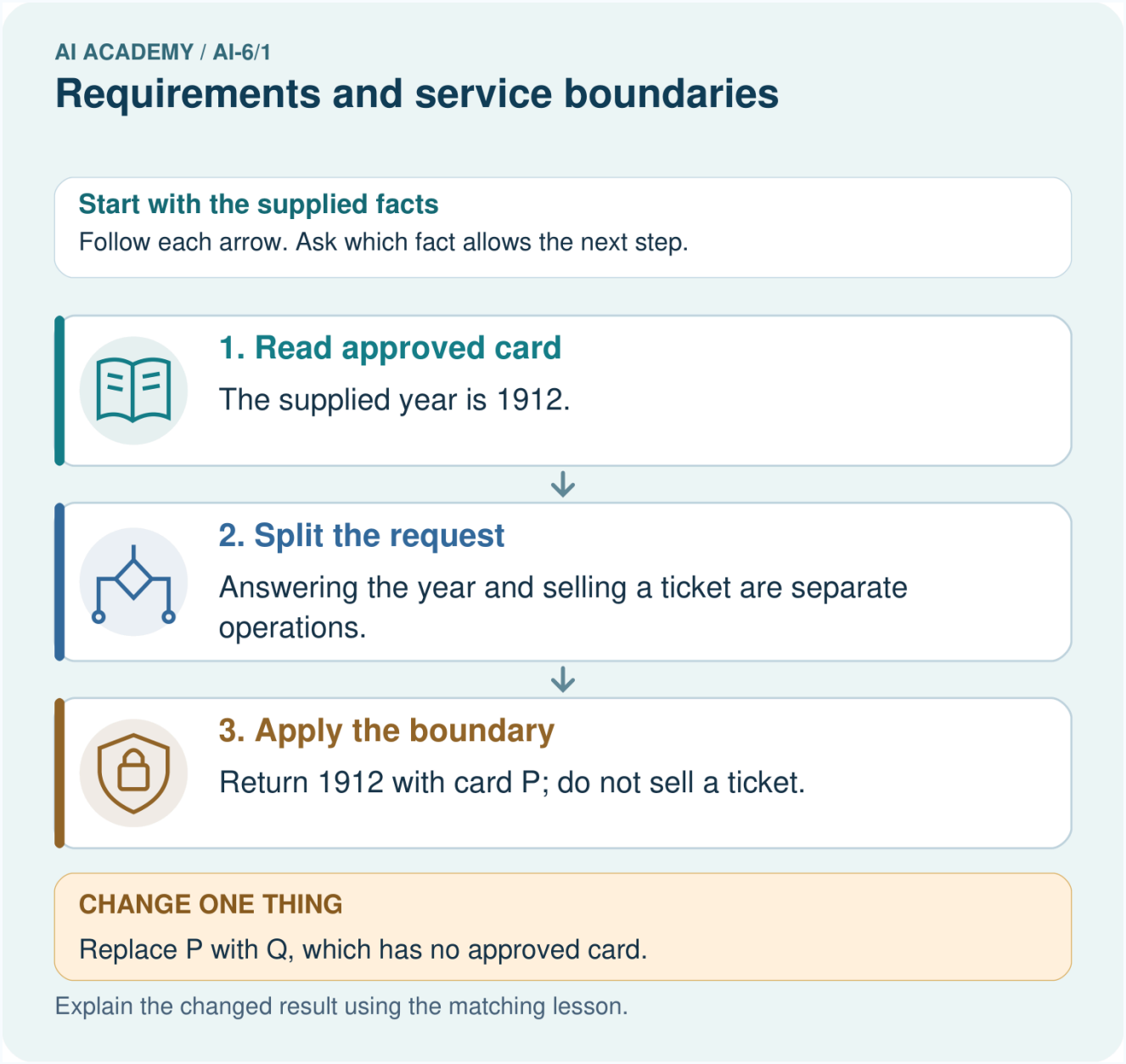
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Requirements and service boundaries

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: A request asks for Tuesday meeting time and to move it to Wednesday. The public schedule lists Tuesday 16:00. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Report Tuesday 16:00 from the schedule; refuse the edit because it is outside the **service boundary**.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Requirements and service boundaries. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A helpful answer does not grant an operation that the service contract excludes.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Split the museum visitor's request into two operations. Which can the service perform for P, and what happens to each operation when Q has no approved card?

The supplied year is 1912. Answering the year and selling a ticket are separate operations. Return 1912 with card P; do not sell a ticket. Changed case: The year is unsupported, and the ticket sale is still excluded.

Knowledge and permission are separate requirements.

Changed-case reasoning: The year is unsupported, and the ticket sale is still excluded. Knowledge and permission are separate requirements.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main mixed request: answer Tuesday16:00 with supplied schedule source; reject move; keep schedule unchanged. Wednesday read reviews, email excludes, missing day rejects invalid input. Museum P1912 is source-supported, sale excludes; Q year unsupported and sale still excluded. Independent inventory: report A available from supplied inventory; do not delete B.

Require separate operation decisions and no invented execution or human review outcome.

Diagnose the current misconceptions

The worked result establishes every future case: A helpful answer does not grant an operation that the service contract excludes.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: A read-only inventory service is asked to list item A and delete item B. Inventory says A is available. Trace permitted work.

Teacher explanation: Report A availability using the source; deletion is outside scope and must not execute.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: A **service boundary** states who can request what, what data may be read and which actions are outside scope. Requirements must be observable.

The fictional student timetable service reads a public schedule and returns a meeting time or review status. It cannot edit schedules or email users. Test success and refusal before adding model behaviour.

W2 Trace the worked case

A request asks for Tuesday meeting time and to move it to Wednesday. The public schedule lists Tuesday 16:00. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Report Tuesday 16:00 from the schedule; refuse the edit because it is outside the **service boundary**.

W3 Audit the limit

A report omits this qualification: A helpful answer does not grant an operation that the service contract excludes. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A helpful answer does not grant an operation that the service contract excludes.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A helpful answer does not grant an operation that the service contract excludes.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: Report Tuesday 16:00 from the schedule; refuse the edit because it is outside the **service boundary**. Qualification: A helpful answer does not grant an operation that the service contract excludes.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

A read-only inventory service is asked to list item A and delete item B. Inventory says A is available. Trace permitted work.

Teacher explanation: Report A availability using the source; deletion is outside scope and must not execute.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Report A availability using the source; deletion is outside scope and must not execute. Require intermediate reasoning and the stated scope.

Fresh retry: Read-only room service is asked to report capacity 8 and book the room. Expected reasoning: Reporting may proceed from its source; booking is denied by the declared boundary.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

Add a labelled requirements and service boundaries **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of requirements and service boundaries.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Architecture and data flow. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/2, ai-6/4, ai-6/7. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of requirements and service boundaries

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: A request asks for Tuesday meeting time and to move it to Wednesday. The public schedule lists Tuesday 16:00. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Report Tuesday 16:00 from the schedule; refuse the edit because it is outside the **service boundary**.

Check every learner before discussion: Use the worked case for Requirements and service boundaries. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A helpful answer does not grant an operation that the service contract excludes.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Replace P with Q, which has no approved card. Require a **prediction** and a reason before comparing. Expected result: The year is unsupported, and the ticket sale is still excluded.

Reason: Knowledge and permission are separate requirements.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

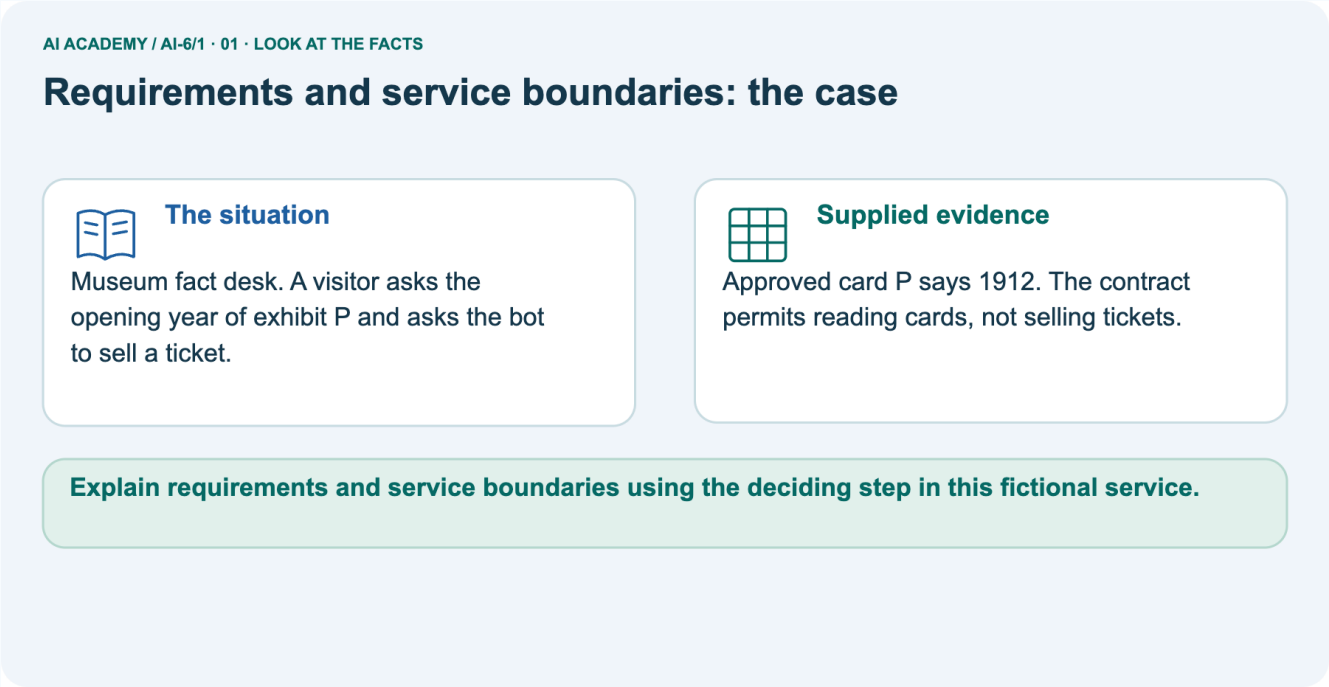
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Requirements and service boundaries: the case



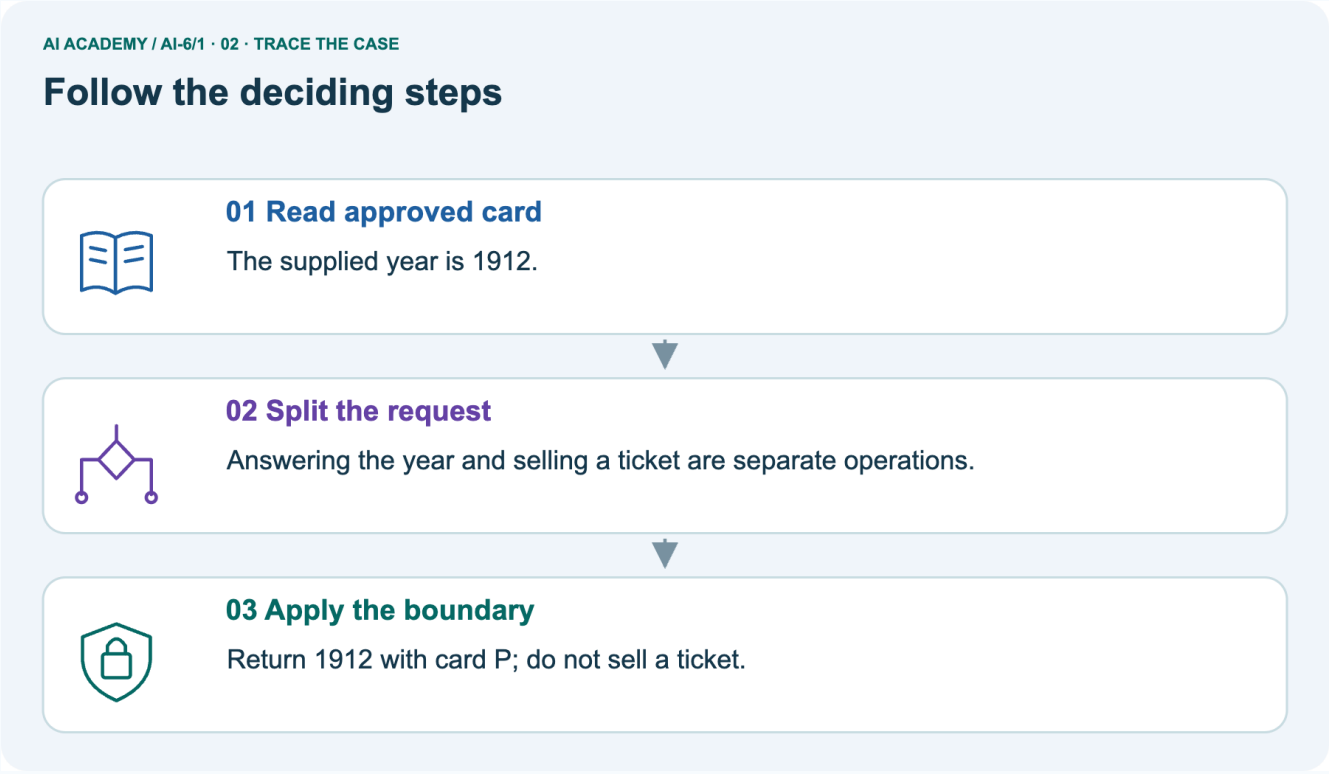
Follow the labelled picture with the matching explanation. Open the vector diagram

Museum fact desk. A visitor asks the opening year of exhibit P and asks the bot to sell a ticket. Supplied facts: Approved card P says 1912. The contract permits reading cards, not selling tickets.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



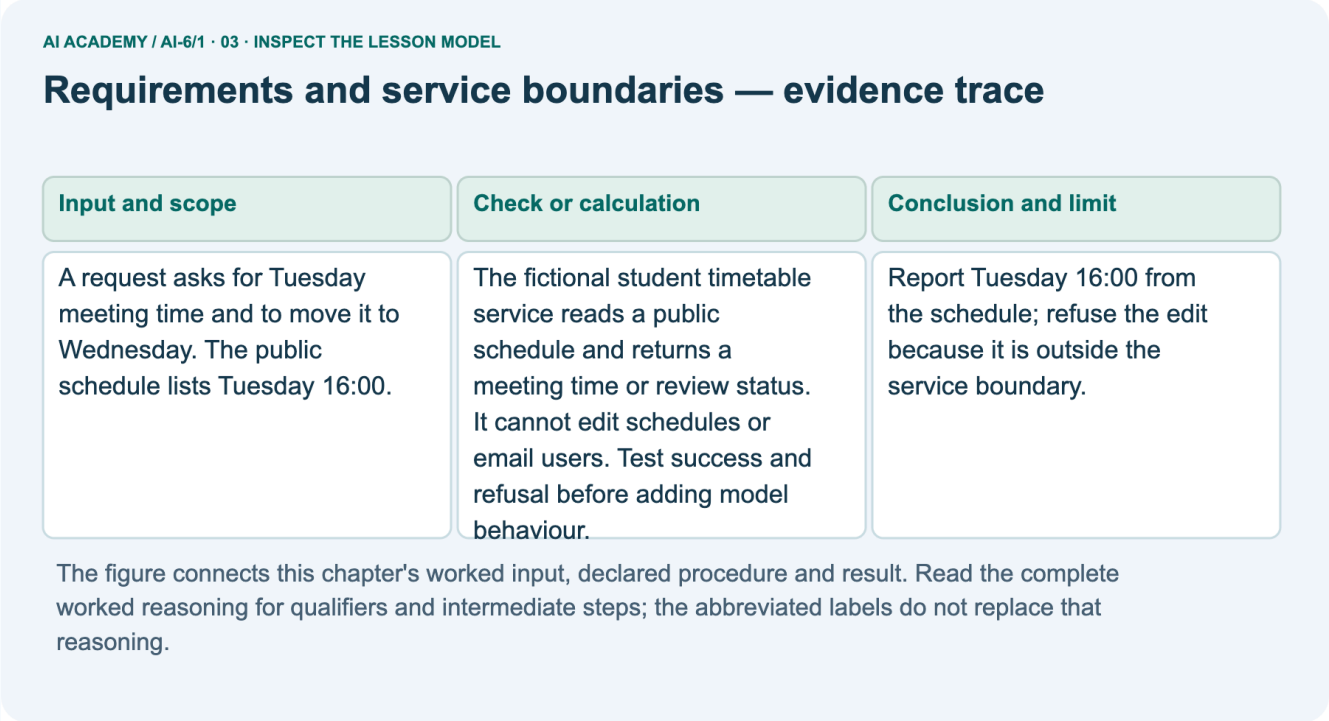
Follow the labelled picture with the matching explanation. Open the vector diagram

Read approved card: The supplied year is 1912. → Split the request: Answering the year and selling a ticket are separate operations. → Apply the boundary: Return 1912 with card P; do not sell a ticket.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Requirements and service boundaries — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Requirements and service boundaries — **evidence** trace. A request asks for Tuesday meeting time and to move it to Wednesday.

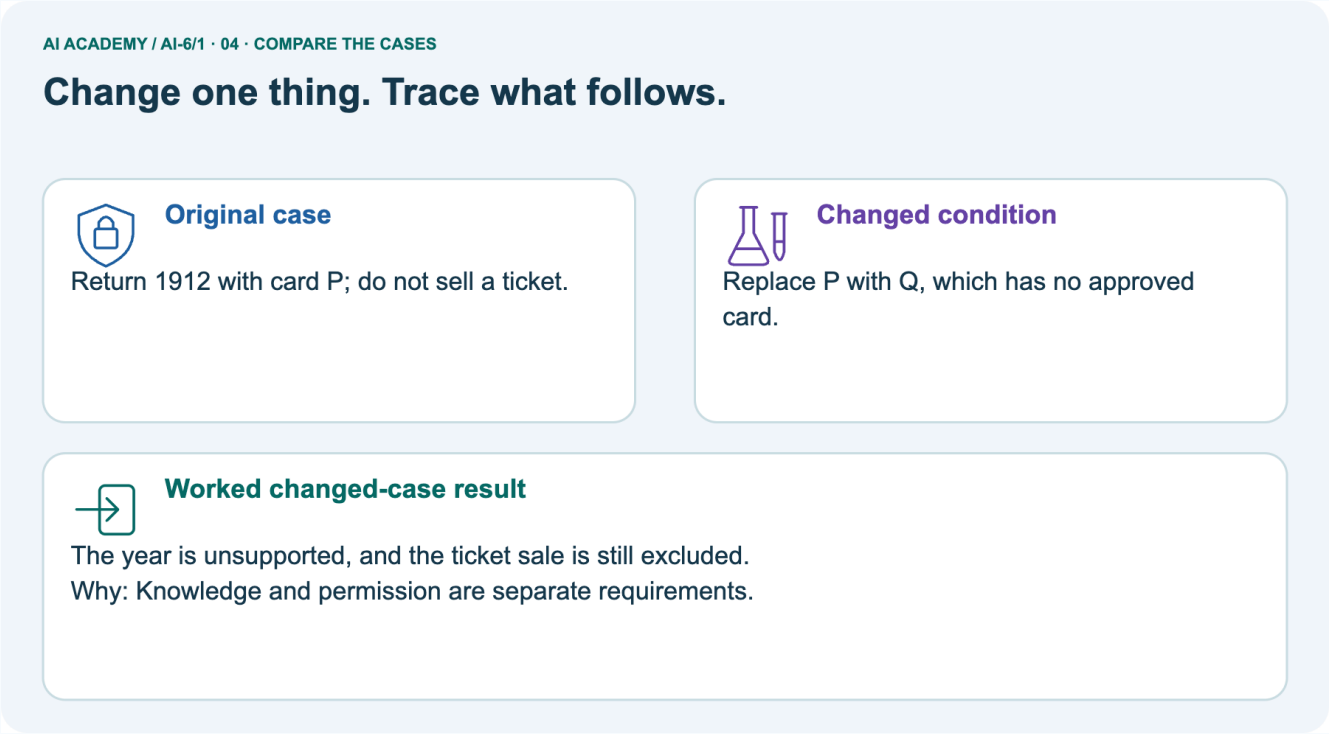
The public schedule lists Tuesday 16:00. | The fictional student timetable service reads a public schedule and returns a meeting time or review status. It cannot edit schedules or email users.

Test success and refusal before adding model behaviour. | Report Tuesday 16:00 from the schedule; refuse the edit because it is outside the **service boundary**.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Return 1912 with card P; do not sell a ticket. Changed condition: Replace P with Q, which has no approved card. Expected result: The year is unsupported, and the ticket sale is still excluded.

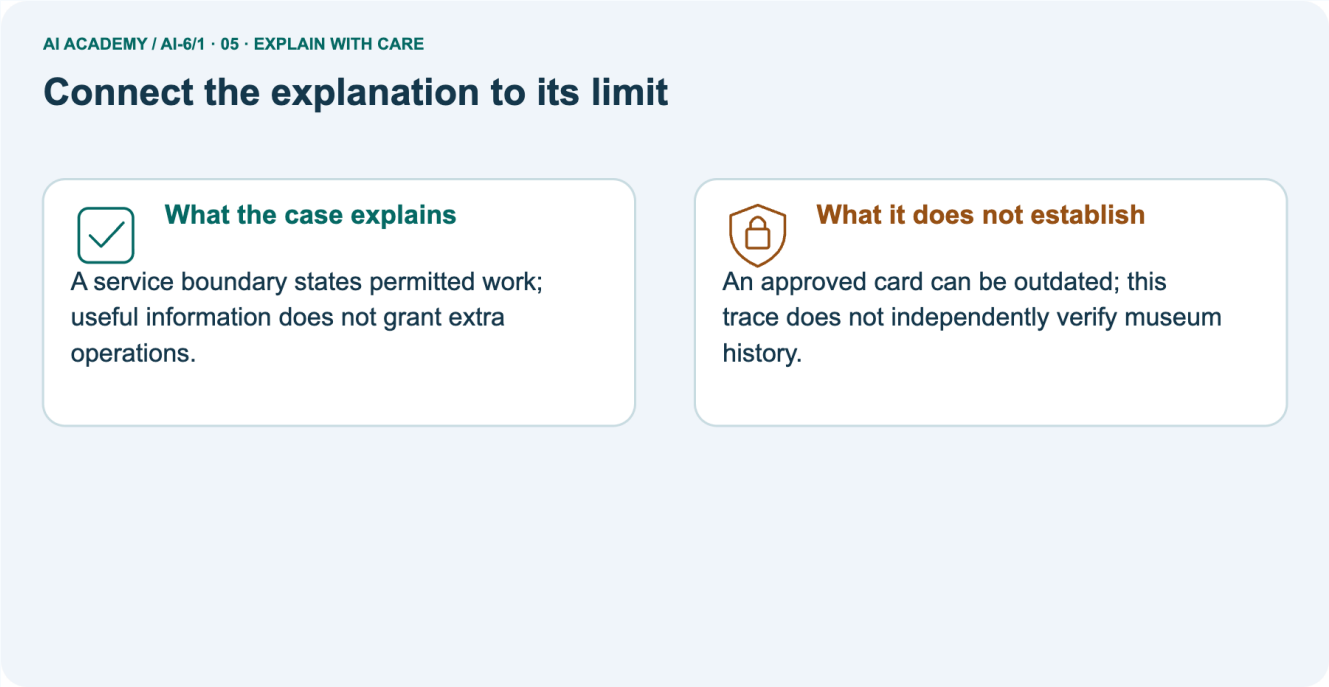
Reason: Knowledge and permission are separate requirements.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Knowledge and permission are separate requirements.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

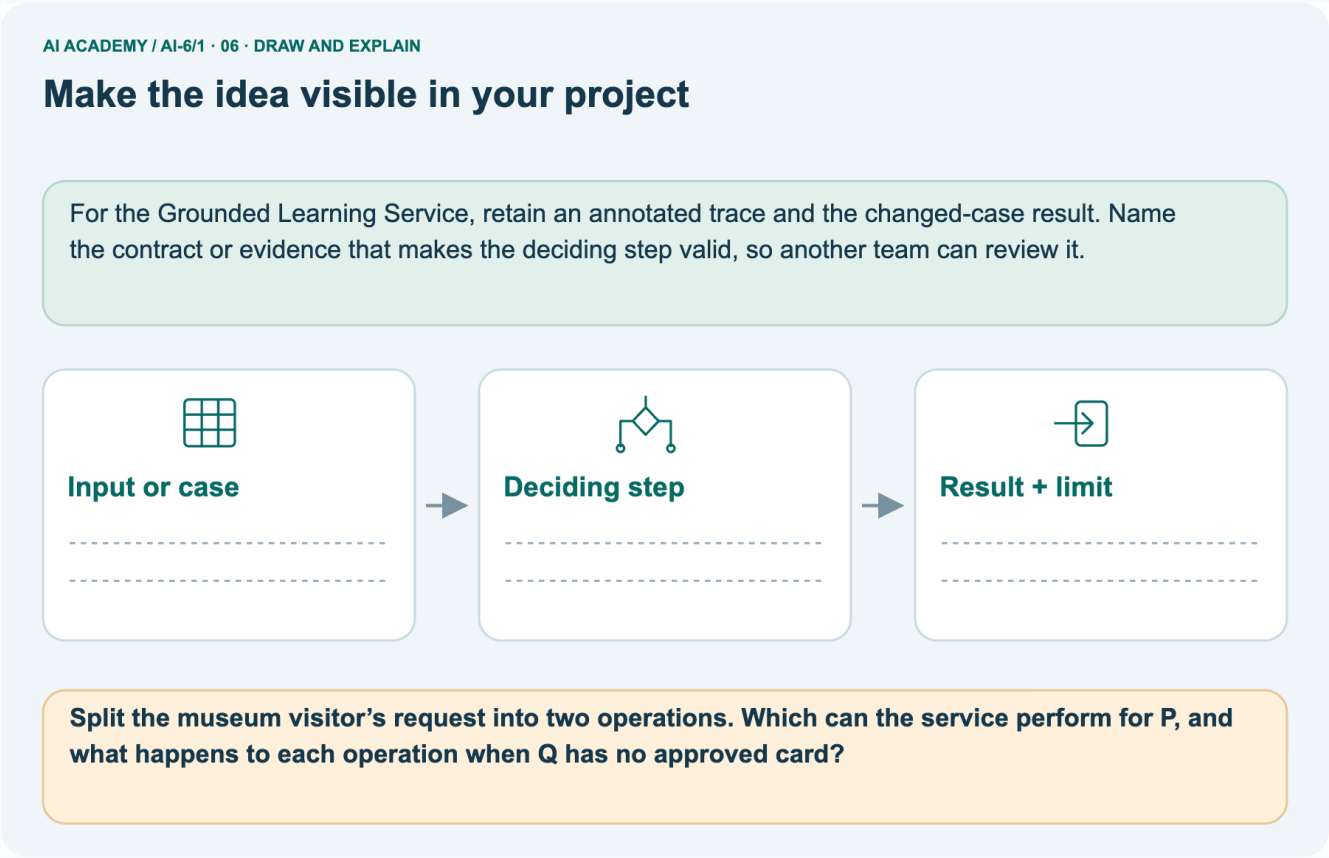
A **service boundary** states permitted work; useful information does not grant extra operations. Limit: An approved card can be outdated; this trace does not independently verify museum history.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: An approved card can be outdated; this trace does not independently verify museum history.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Split the museum visitor's request into two operations. Which can the service perform for P, and what happens to each operation when Q has no approved card?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The supplied year is 1912. Answering the year and selling a ticket are separate operations. Return 1912 with card P; do not sell a ticket.

Changed case: The year is unsupported, and the ticket sale is still excluded. Knowledge and permission are separate requirements.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Requirements and service boundaries: the case

02. Follow the deciding steps

03. Requirements and service boundaries — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.